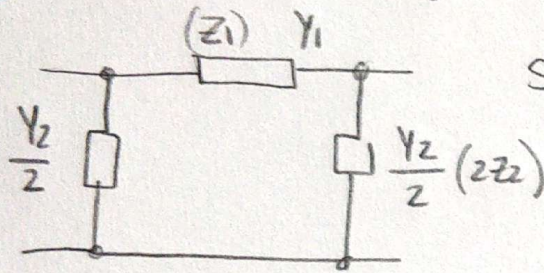


2) Parâmetros imagem π para H



Sistema simétrico

$$\rightarrow Y_1 = \frac{1}{Z_1}$$

$$\rightarrow Y_2 = \frac{1}{Z_2}$$

$$\rightarrow Y_{0\pi} = \sqrt{\Delta Y}$$

$$Y = \begin{bmatrix} Y_A + Y_B & -Y_B \\ -Y_B & Y_B + Y_A \end{bmatrix} = \begin{bmatrix} \frac{Y_2}{2} + Y_1 & -Y_1 \\ -Y_1 & \frac{Y_2}{2} + Y_1 \end{bmatrix}$$

$$\Delta Y = Y_{11} \cdot Y_{22} - Y_{21} \cdot Y_{12} = \left(\frac{Y_2}{2} + Y_1\right)^2 - (-Y_1)^2$$

$$\Delta Y = \frac{Y_2^2}{4} + \frac{Y_2 Y_1}{2} + Y_1^2 - Y_1^2 = \frac{Y_2^2}{4} + Y_2 Y_1$$

$$Y_{0\pi} = \sqrt{\frac{Y_2^2}{4} + Y_2 Y_1} = \sqrt{Y_2 Y_1 \left(\frac{Y_2}{4 Y_1} + 1\right)} = \sqrt{Y_2 Y_1 \left(\frac{Y_2}{4 Y_1} + 1\right)} = \sqrt{Y_2 Y_1} \sqrt{\frac{Y_2}{4 Y_1} + 1} = \sqrt{\frac{C}{L}} \sqrt{\frac{1}{4 g^2 C L} + 1}$$

$$Z_{0\pi} = \frac{1}{\sqrt{Y_2 Y_1} \sqrt{\frac{Y_2}{4 Y_1} + 1}} = \sqrt{\frac{L}{C}} \cdot \sqrt{\frac{4 g^2 C L}{1 + 4 g^2 C L}}$$

$$\rightarrow Z_{s1} = \sqrt{\frac{AB}{CD}} = \sqrt{\frac{Z_{11}}{Y_{11}}} = \sqrt{\frac{\frac{Y_{22}}{\Delta Y}}{Y_{11}}} = \sqrt{\frac{\frac{\frac{Y_2}{2} + Y_1}{\frac{Y_2^2}{4} + Y_2 Y_1}}{\frac{Y_2}{2} + Y_1}} = \sqrt{\frac{\frac{Y_2}{2} + Y_1}{\left(\frac{Y_2^2}{4} + Y_2 Y_1\right) \frac{Y_2}{2} + Y_1}} = \sqrt{\frac{1}{\frac{Y_2^2}{4} + Y_2 Y_1}} = Z_{0\pi}$$

$$\rightarrow Z_{s2} = Z_{s1} = Z_{0\pi}$$